An exercise regime for a retired adult, a 2000-word case report

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This case study focuses on a group of recently retired people, aged around 65 years old. This group has a lethargic lifestyle and has had little or no physical activity in their previous lives. The objective of the case study is to integrate the previous research with the physiological characteristics of this group to design a reasonable exercise programme that can effectively improve functional independence and reduce the risk of falls in this group. It should be noted that the title of the case pointed out that this group has two main characteristics, the first is that this group has no past medical history, which means that my exercise programme does not need to consider whether the target group's past illnesses will affect the selection of the exercise programme or need to give extra consideration to adjusting the intensity of the exercise to stabilise the exercise effect of the expectations; the second is that the target group is worried about the psychological state of starting to exercise. Then, how to make the target population enjoy participating in exercise (which is equivalent to giving the target population the right motivation and incentive to continue exercising to achieve results) will be one of the issues that I need to consider.

According to the World Health Organisation (WHO, 2023), people aged 65 years and over are usually classified as older adults. This age group has an increased risk of loss of functional independence and a higher prevalence of age-related diseases (Valenzuela et al.) Falls are common among older adults and can cause immediate and long-term physical trauma (Mohr et al., 2023). Therefore, preventive interventions aimed at increasing functional independence and reducing the incidence of falls have become increasingly important, especially considering changing demographics.

According to the Functional Independence Measurement Scale (FIMS) (Physiopedia contributors, 2024), a normal, intact, independent person has 18 independent functions, thirteen of which are physical-motor tests and five of which are cognitive tests (Shirley Ryan Ability Lab, 2025). Categorised these independent functions of the body according to whether an exercise programme could directly or indirectly enhance them, and such categorisation is a tailored intervention that effectively prevents declines in functional independence to the target group (Molenaar et al., 2022). Cognitive comprehension Expression, social interaction action, problem-solving, and memory are unquestionably the cognitive categories of independence that are indirectly or marginally improved by exercise and will not be the focus of the exercise programme. Meanwhile, eating (Miyauchi et al., 2024) (Melo et al., 2021), grooming (Fujita et al., 2017) (Mlinac et al., 2016) (Melo et al., 2021), bathing (Melo et al., 2021) (Khayatzadeh-Mahani et al., 2023) (Mlinac et al., 2016), Upper body dressing (Melo et al., 2021) (Christie et al., 2011), Lower body dressing (Melo et al., 2021) (Christie et al., 2011), toileting (Talley et al., 2014), Bladder management (Mizrahi et al., 2010), Bowel management (Potter et al., 2005), Bed to chair transfer (Physiopedia, 2024), toilet transfer (Grimsland et al., 2019), shower transfer (De-Rosende-Celeiro et al., 2019), locomotion (Jung, Hungu et al., 2021) (Sá et al., 2019), and going up and down stairs (Gagliano-Jucá et al., 2020) （Tiedeman et al., 2007） are independent functions that can be directly improved by exercise. These aspects will be analysed and improved by exercise.

After separating the independent functions into groups, exercise schedules can be created by isolating single motor behaviours while combining the same ones (Koyama et al., 2006). In this process of analysis, extraction, combination, design, and exercise, the independent functions of the human body are targeted at a single motor behaviour rather than a combination of multiple motor behaviours, feedback is needed to set up the results of the training in the subsequent exercise programme to determine the effectiveness and improvement of the exercise (Weakley et al., 2023).

The complete eating process can be divided into four stages: reaching, spooning, transporting, and chewing (Nakatake et al., 2021). The independent functions can be summarised as flexion and extension of the upper limbs, grasping objects by hand (fingers) and using the masticatory muscles. The complete grooming process involves the use of grooming tools, such as combs, toothbrushes, razors, and the removal, cleaning, and reinstallation of dental orthotics and dentures, which require precise control of the hand muscles (American Occupational Therapy Association, 2014). Considering that most of the time in grooming is standing, it is also a priority to enhance muscle groups that contribute to the ability to stand with Balance and Proprioception, such as the Quadriceps, Hamstrings, Erector spinae, Calves, and abdominal muscles (Kim et al., 2020). Bathing involves precise control of the hand muscles and the ability to grasp objects, as well as upper and lower extremity muscle strength and abdominal core strength to maintain body stance during bathing. The table below presents the remaining ten independent functions and their associated single motor behaviours (Kane, 2020).

independence function	motor behavior				
Eating	flexion and extension of the upper limbs	grasping of objects by the hand (fingers)	use of the masticatory muscles		
Grooming	precise control of hand muscles	low limb muscle	abdominal muscle		
Bathing	flexion and extension of the upper limbs	Precise control of hand muscles	upper limb muscle	low limb muscle	abdominal muscle
Upper Body Dressing	shoulder and arm mobility	Precise control of hand muscles			
Lower Body Dressing	trunk stability	lower extremity coordination	hand grip for balance		
Toileting	squatting	trunk stability	hand-eye coordination		
Bladder Management	precise control of hand muscles				
Bowel Management	postural stability	abdominal muscle			
Bed-to-Chair Transfer	abdominal muscle	coordinated weight shifting.	low limb muscle		
Toilet Transfer	abdominal muscle	low limb muscle			
Shower Transfer	abdominal muscle	low limb muscle			
Locomotion	coordination of lower extremities	Gait	low limb muscle	abdominal muscle	
going up and down stairs	Gait	low limb muscle	abdominal muscle	upper limb muscle	

Based on the table's contents, another count was made, as presented in the table below, of the number and frequency of occurrences of the single motor actions around which the movement design should be centred in enhancing the individual's independent functionality.

abdominal muscle	0.216216216	8
low limb muscle	0.189189189	7
precise control of hand muscles	0.108108108	4
trunk stability	0.054054054	2
upper limb muscle	0.054054054	2
Gait	0.054054054	2
flexion and extension of the upper limbs	0.054054054	2
grasping of objects by the hand (fingers)	0.027027027	1
shoulder and arm mobility	0.027027027	1
hand-eye coordination	0.027027027	1
use of the masticatory muscles	0.027027027	1
squatting	0.027027027	1
coordinated weight shifting	0.027027027	1
hand grip for balance	0.027027027	1
postural stability	0.027027027	1
coordination of lower extremities	0.027027027	1
lower extremity coordination	0.027027027	1

The exercise training reported in this case study also places Balance and Proprioception at the centre of the training. Ageing leads to varying degrees of degeneration and atrophy of both neural pathways and muscles, which can significantly reduce an individual's ability to balance and proprioception (Henry et al., 2019). At the same time, the reduction in balance and proprioception ability can greatly increase the risk of falls in the elderly population (Wang et al., 2022). So, improving the ability to go for balance and proprioception at the primary geriatric stage, i.e., age 65 years where the target individuals are located, is an effective measure to prevent the increased risk of falls in the later geriatric stage.

The analysis of an exercise programme for balance and proprioception ability still begins with the abstraction of a single factor for both. An individual's state of balance is generally divided into dynamic and static balance, and both need to be considered when designing an exercise programme. Static balance is mainly related to trunk extensors (Golubić et al., 2021), trunk muscles (Acar et al., 2020), ligaments and muscle groups associated with the ankle, knee, and hip (Babayi et al., 2023), while dynamic balance is related to cardiorespiratory fitness (Jiroumaru et al., 2023), lower extremity strength, trunk extensors (Golubić et al., 2023), Ankle, knee and hip-related ligaments and muscle groups (Babayi et al., 2023). For proprioception, on the other hand, whole-body neural pathway sensitivity needs to be strengthened, along with increasing whole-body muscle strength to prevent muscle ageing and degradation leading to decreased neural pathway sensitivity (Henry et al., 2019). During research on how to prevent falls, it has been found that the more internally confident the older age group older than 65 is, the lower the risk of falls (Litwin et al., 2018). This has strong relevance to our research population. The illustration in the case provided to the fact that our study subject had a worried mindset about starting exercise suggests that establishing an exercise training programme needs to be considered in terms of building exercise confidence, giving motivation to the exercising individual and removing the mental aspects of negative worry and negativity.

After separating the core objectives of the case requirements, we can target training planning to a single motor behaviour. Under the umbrella of planning training, there are three principles of exercise programme planning to be considered: FITT (frequency, intensity, time, type), SAID (specific, adaptation, imposed, demands), and MAP (motivation, ability, performance). Because the physiological state of the participant declines over time as the trainer participates in the training programme（Valenzuela et al. 2019）, the training programme should be set up as a circuit training programme so that after building the participant's confidence in exercise, the participant can exercise every day according to the training programme. Based on the above information, we can divide the exercise programme into three phases: the beginning phase, the intensity increase phase, and the circuit training phase. The main purpose of the beginning stage is to cultivate participants‘ confidence in exercise and make psychological preparation for the intensity increase stage; the intensity increase stage mainly focuses on increasing the intensity of exercise in a step-wise manner, accompanied by the variation of exercise, to avoid participants' loss of interest, and at the same time, to make physiological preparation for the training in the cyclic stage; the cyclic stage mainly aims to cultivate the participant's exercise habits, and to effectively improve the participants' functional independence and balance as well as their proprioception. The cycling phase is mainly to cultivate participants' exercise habits and effectively improve their functional independence, balance and proprioception.

The duration of the beginning phase of the exercise programme was set at two weeks, with the main exercise being slow walking in any location, suggesting a park, or the option of overlapping this exercise with daily purchasing behaviour, such as walking on the way to the supermarket as a strolling exercise content endurance exercise for older people. According to the ACSM (2009) Position Statement on Exercise and Physical Activity for Older Adults, it is mentioned that the frequency of slow walking will be set at a minimum of 10 minutes per session for at least 30 minutes per day for the first week, and 60 minutes per day for the second week, for a total of approximately 150-300 minutes per week. Participants will be asked to socialise while doing the slow walking exercise, whether it is socialising with other exercisers while walking in the park, playing chess with others while taking a break at the end of a slow walk in the park, or chatting to the cashier after a supermarket purchase. This requirement was primarily designed to help participants expand their social space, and on the one hand, maintaining such a social construct required participants to consistently complete the exercise programme i.e., it gave participants the motivation to keep exercising to support social interaction with others, and at the same time, it increased participants' confidence in exercising and eliminated the exerciser's psychological concerns. Muscle stretching and relaxation after the slow walk and a more detailed programme of adjustments for the first phase can be found in the Appendix.

The intensity increase phase of the workout begins with the inclusion of a workout programme targeting higher frequency single exercise movements, which again is divided into two weeks with a gradual increase in training. To ensure the effectiveness of our training, the intensity of the workout needs to be 60% to 85% of the individual's maximum strength, and healthy older adults should train three to four times per week (Mayer et al., 2011). At this stage, the choice is to include in the training programme exercises that target the lifting of the abdominal muscle, low limb muscle, precise hand control, trunk extensors, trunk muscles, ligaments and muscle groups associated with the ankle, knee and hip joints. For the abdominal muscles, a good core program should target multiple abdominal core groups, so the three movements of the bridge, modified planks, and opposite arm and leg raises would be used (Harvard Health Publishing 2020). For the low limb muscles, choose standing Marches, seated knee extensions, and standing heel raises; trunk extensions would be back extensions and cat-cow stretches; and trunk muscles would be planks and side planks, which were combined because they overlapped with those of the abdominal muscles. The ankle exercises were seated-ankle circles, toe taps, and standing heel raises; the knee exercises were leg extensions, knee flexion and extension, and squats; and the hip exercises were hip abductions, hip adductions, and hip adductions. Hip abductions, hip adductions, bridges (this item overlaps with the abdominal muscle training). The objects to be grasped will be selected to improve the grasping ability and accuracy of the hand, with priority given to recreational items such as mahjong, poker and chess, which have social attributes, so that the subject can improve his hand ability by grasping the pieces of mahjong, poker and chess, If there is no opportunity for the above recreational activities to occur, the training participant may choose to use chopsticks instead of the training content（Yildirim et al., 2021）. The detailed training plan for this phase is placed in the Appendix.

The cycling phase of the exercise programme will continue with the last training session of the Intensity Enhancement Phase, and two new items will be added during this phase. Firstly, a feedback mechanism will be set up to determine the usefulness of the training programme and the appropriateness of the intensity of the training through telephone interviews with the trainees, offline observations, and self-confirmation by the trainees（Weakley et al., 2023）. This feedback mechanism will start on the second Sunday of the second week of the cycle phase of the programme and will be conducted every fortnight to facilitate the adjustment of the training content. Secondly, the whole-body training programme, Tai Chi and Yoga, was increased to improve cardiorespiratory fitness, balance and flexibility while reducing the risk of falls (Chen W et al., 2023) (Hao, Zikang et al., 2022). The specific training schedule is shown in the appendix.

Overall, this case report presents a phased exercise programme designed to enhance functional independence and reduce the risk of falls in a recently retired, exercise-poor older population aged around 65 years. By combining the principles of FITT, SAID and MAP, the programme focuses on balance and proprioceptive training, and gradually enhances the participants' exercise capacity in three stages: from the slow walking stage to build up the exercise habit, to the intensity incremental training targeting the key muscle groups, and finally to achieve the long term health goal through circuit training and whole-body exercise such as tai chi and yoga. The programme pays special attention to psychological motivation, such as social interaction and feedback mechanisms, to ease older people's concerns about exercise.

Total word number：2335

Appendix

beginning phase

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 1	slow walking(T1)	slow walking(T1)	slow walking(T1)	slow walking(T1)	slow walking(T1)	slow walking(T1)	slow walking(T1)
Week 2	slow walking(T2)	slow walking(T2)	slow walking(T2)	slow walking(T2)	slow walking(T2)	slow walking(T2)	slow walking(T2)

Slow walking(T1) refers to type one slow walking, i.e., the frequency is 10 to 15 minutes, a total of 30 minutes throughout the day, and the time arrangement is suggested to be once after breakfast and once before lunch, and once before or after dinner, which are more likely to occur by the social requirements of the participants as mentioned in the text. Exercise Adjustment Suggestion: If after 15 minutes of slow walking, the body still feels good, can increase the time of slow walking to 30 minutes to an hour at a time if can't judge the physical state, try to have a social act after the required 15 minutes of slow walking if the participant can carry on a complete, uninterrupted dialogue, it means that you can choose to carry on the slow walking exercise for 30 minutes to an hour. one hour of slow walking exercise.

Slow walking (T2) refers to type two slow walking, i.e., 60 minutes per day, 150 to 300 minutes per week, with the same duration recommendations as type one slow walking, with the same intensity adjustments, and, if physically possible, increasing the duration of slow walking to 30 minutes, with the same adjusted self-judgment conditions as type one slow walking.

Saturday and Sunday workouts can be chosen freely in this beginning phase of the workout. At the end of each workout, a post-walking stretching session was performed to stretch the muscles, increase flexibility, reduce stiffness, and improve blood circulation in the anterior and posterior calf muscles. The entire muscle stretching process is set for five to ten minutes. The entire stretching process is aimed at stretching and relaxing the anterior, posterior, and anterior and posterior groups of the calves at the same time.

Torso Bend Forward Stretch
Action:
1. Stand with legs shoulder-width apart and your hands hanging naturally.
2. Slowly bend forward, try to touch the ground with both hands and keep your knees slightly bent.
3. Stay when you feel a slight stretch in the muscles at the back of your legs.
4. Hold for 15-30 seconds.
Repeat: 2-3 times.

Stretching needs to be done slowly to avoid strains caused by pulling hard or pushing too hard. Stretching should be done with a slight sensation of stretching, and if pain occurs, you need to stop the current stretching movement to avoid injury. The time and number of stretches should be increased gradually according to your situation to avoid over-exercise.

intensity increase stage

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 3	slow walking(T3) Circuit 1 2 x a day	slow walking(T3) Circuit 1 2 x a day	slow walking(T3) Circuit 2 2 x a day	slow walking(T3) Circuit 2 2 x a day	slow walking(T3) Circuit 2 2 x a day	slow walking(T3)	slow walking(T3)
Week 4	slow walking(T3) Circuit 3 2 x a day	slow walking(T3) Circuit 3 2 x a day	slow walking(T3) Circuit 3 2 x a day	slow walking(T3) Circuit 4 2 x a day	slow walking(T3) Circuit 4 2 x a day	slow walking(T3)	slow walking(T3)

During the period of increased intensity, slow walking (T3) will still be included in the workout programme, but the duration of the workout will be adjusted to only one session a day lasting 45 minutes to 1 hour, with the yellow Saturday and Sunday still available to you depending on your fitness level. This adjustment was made because of the need to free up time and fitness for other workouts for the participants. In addition to the same workouts, the workouts for this phase are bridge, modified planks, opposite arm and leg raises, and side planks.
Back extensions, cat-cow stretches， seated-ankle circles, toe taps, standing heel raises, leg extensions, knee flexion and extension, squats, hip abductions, hip adduction
The workouts are organised according to a gradual increase in intensity and are divided into Circuit 1, Circuit 2, Circuit 3, and Circuit 4. Apart from the circuits, maintain social recreational activities such as chess, poker, etc. during slow walks to maintain and improve the hand capacity of the training target.

Circuit 1

Exercise	Repetitions
Bridge	10 reps per set, 2 sets
Modified Planks	Hold for 10-15 seconds per set, 2 sets
Cat-Cow Stretch	5 repetitions per movement, 2 sets
Seated Ankle Circles	5 circles per foot, 2 sets

Circuit 2

Exercise	Repetitions
Bridge	12 reps per set, 2 sets
Modified Planks	Hold for 20 seconds per set, 2 sets
Opposite Arm and Leg Raise	8 reps per side, 2 sets
Side Plank	Hold for 10 seconds per side, 2 sets
Toe Taps	10 reps per set, 2 sets

Circuit 3

Exercise	Repetitions
Bridge	15 reps per set, 3 sets
Modified Planks	Hold for 30 seconds per set, 3 sets
Opposite Arm and Leg Raise	10 reps per side, 3 sets
Side Plank	Hold for 20 seconds per side, 3 sets
Seated Ankle Circles	10 circles per foot, 3 sets
Standing Heel Raises	10 reps per set, 3 sets

Circuit 4

Exercise	Repetitions
Bridge	20 reps per set, 3 sets
Modified Planks	Hold for 40 seconds per set, 3 sets
Opposite Arm and Leg Raise	12 reps per side, 3 sets
Side Plank	Hold for 30 seconds per side, 3 sets
Toe Taps	15 reps per set, 3 sets
Standing Heel Raises	15 reps per set, 3 sets
Leg Extensions	10 reps per side, 2 sets
Knee Flexion and Extension	10 reps per set, 2 sets
Squats	10 reps per set, 2 sets

Explanation of exercises in Circuit 1，2，3，4：

Bridge：

Lie on your back with your knees bent and feet flat on the floor, shoulder-width apart, and arms by your sides. Tighten your abdominal and glute muscles, then lift your hips toward the ceiling, creating a straight line from shoulders to knees. Hold briefly, and slowly lower your hips back to the floor.

Modified Planks
Start on your hands and knees, keeping your shoulders over your wrists and your hips over your knees. Tighten your core and extend one leg behind you, followed by the other, keeping your body straight. Hold this position while engaging your core, and if necessary, drop to your knees for support.

Opposite Arm and Leg Raise,
Start on all fours, keeping your hands under your shoulders and your knees under your hips. Slowly raise your right arm in front of you and your left leg behind you, keeping both straight. Hold briefly, then return to the starting position and repeat with the left arm and right leg.

Side Plank
Lie on your side with your legs stacked, elbow beneath your shoulder, and body in a straight line. Lift your hips off the floor, supporting yourself on your forearm and the edge of your bottom foot. Hold the position for a few seconds, then repeat on the other side.

Back Extensions
Lie face down on the floor with your arms extended in front of you. Slowly lift your chest and upper body off the ground, engaging your lower back and glutes. Hold briefly at the top, then lower back down.

Cat-Cow Stretch
Start on all fours with your hands under your shoulders and knees under your hips. Inhale as you arch your back, lifting your head and tailbone (cow position). Exhale as you round your back, tucking your chin and drawing your tailbone under (cat position). Repeat for the prescribed repetitions.

Seated Ankle Circles
Sit in a chair with feet flat on the floor. Lift one foot off the ground and rotate your ankle in circles, first in one direction, then the other. Repeat with the other foot.

Toe Taps
Stand with your feet hip-width apart and knees slightly bent. Lift your right foot and gently tap your toes on the floor before you, then quickly return to the starting position. Alternate tapping with the left foot.

Standing Heel Raises
Stand tall with feet hip-width apart, holding onto a chair or wall for support. Slowly raise your heels off the ground, standing on your toes. Hold briefly, then lower your heels back to the floor.

Leg Extensions
Sit in a chair with your feet flat on the floor and knees bent at 90 degrees. Slowly extend one leg out in front of you, straightening it completely. Hold for a moment, then lower it back to the floor. Repeat on the other side.

Knee Flexion and Extension
Sit in a chair with feet flat on the floor and knees at a 90-degree angle. Slowly straighten one leg in front of you (extension), then bend the knee back to the starting position (flexion). Repeat with the other leg.

Squats
Stand with feet shoulder-width apart and toes pointing slightly outward. Bend your knees and lower your hips as if sitting back into a chair, keeping your chest upright. Lower until your thighs are parallel to the floor, then push through your heels to return to the starting position.

Hip Abductions
Stand tall with feet hip-width apart, holding onto a chair or wall for support. Slowly lift your right leg out to the side, keeping it straight. Hold briefly at the top, then lower it back down. Repeat on the other leg.

Hip Adduction
Stand with feet hip-width apart, holding onto a chair or wall for support. Slowly bring your right leg towards your left leg, keeping it straight. Hold briefly, then return to the starting position. Repeat with the other leg.

cycling stage

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 3	slow walking(T3) Circuit 4 2 x a day Yoga	slow walking(T3) Circuit 4 2 x a day Tai chi	slow walking(T3) Circuit 4 2 x a day Yoga	slow walking(T3) Circuit 4 2 x a day Tai chi	slow walking(T3) Circuit 4 2 x a day Yoga	slow walking(T3)	slow walking(T3)
Week 4	slow walking(T3) Circuit 4 2 x a day Yoga	slow walking(T3) Circuit 4 2 x a day Tai chi	slow walking(T3) Circuit 4 2 x a day Yoga	slow walking(T3) Circuit 4 2 x a day Tai chi	slow walking(T3) Circuit 4 2 x a day Yoga	slow walking(T3)	slow walking(T3) first feedback

Yoga & Tai chi： Where possible, these two exercises can be performed outdoors. Considering the need to improve the confidence of the trainee to assist him/her to perform better, the trainee can also choose to perform Yoga and Tai Chi at home to ensure the completeness of the exercise movements and the effect of the exercise.

Feedback：

1. Physical strength and endurance
Gait and stride:
Observe whether the gait is more stable and the stride is more fluid before and after the exercise.

6-minute walk test:
Record how far older adults can walk in 6 minutes to assess their endurance.

Standing Endurance:
Tests how long an older person can stand continuously, or how many times they sit down after standing.

2. Muscle strength
Grip strength test:
Using a grip strength meter to test the strength of the grip of both hands. This is a common measure of upper body strength and overall physical fitness.

Lower Body Strength Test:
For example, the number of times you sit down and stand up to test the strength and endurance of your lower body.

Leg extension:
To test the strength and flexibility of thigh and calf muscles.

3. Flexibility and Balance
Sitting Forward Bend Test:
Tests flexibility, especially the extension of the lower back and the back of the thighs.

One Leg Stand Test:
Tests balance by asking the older person to stand on one leg without support and recording how long they can do so.

TUG (Timed Up and Go):

Older adults get up from a seated position, walk a certain distance, and then return to a seated position to assess speed and stability of movement and reaction.

4. Functional Movement
Stair climbing test:
Tests the older person's leg strength and co-ordination.

Walking speed:
Measures the speed of walking over a certain distance and reflects the mobility of the older person.

Sit-to-Stand Test:
Tests the speed and control of standing from a seated position and assesses the strength of core muscles.

5. Health indicators
Weight, BMI:
Tracking changes in weight and BMI (Body Mass Index) to help assess whether older adults are making improvements in weight management.

Blood Pressure:
Check blood pressure regularly to see if there is any improvement in blood pressure due to exercise.

Heart rate:
Check resting heart rate versus recovery heart rate after exercise to assess heart health.

6. Pain and Functional Improvement
Pain assessment:
Find out if there is any pain relief or aggravation after exercise (especially joint or muscle pain).

Mobility scores:
Ask the older person about their self-assessment of their ability to perform daily activities, such as whether they can carry out daily tasks (e.g., shopping, cleaning, caring for themselves, etc.) with greater ease.

7. Mental Health and Mood
Self-perception score:

Assesses the subject's psychological state before and after exercise, whether they feel more energized, less anxious or depressed.

Quality of Life Assessment:
Ask older adults how they rate their quality of life to see if exercise has improved their overall sense of well-being.

8. Exercise Continuity and Participation
Compliance assessment:
To check whether the participant adheres to the exercise programme and whether he/she stops or reduces the frequency of exercise in the middle of the programme.

Feedback and Suggestions:
Through communication with the participants, to understand their views on the exercise programme and whether they have any difficulties or discomfort, to further optimize the exercise programme.

9. Feedback programme design points:
Composite indicators:
Combine indicators from multiple areas above to assess the overall effectiveness of the exercise programme, not just weight or walking speed.

Personalised Adjustment:
Adjust the exercise programme according to the feedback and assessment results of the participants to ensure that the exercise content meets their health needs and abilities.

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